

TABLE II

SUCCESS RATE (SR, %) ACROSS THE ELEVEN LIBERO-VPRO CHALLENGE CATEGORIES SHARED BY ALL SIX MODELS. NORMAL DENOTES CLEAN-CONDITION PERFORMANCE, WHILE GRAY $\downarrow$ VALUES REPORT THE ABSOLUTE SR DROP RELATIVE TO NORMAL. CATEGORIES MARKED $\uparrow$ ARE ROBUSTNESS INDICATORS, WHERE HIGHER SR INDICATES GREATER RESILIENCE. CATEGORIES MARKED $\diamond$ ARE DIAGNOSTIC INDICATORS, WHERE HIGH SR MAY REFLECT RELIANCE ON NON-VISUAL CUES OR LEARNED PRIORS RATHER THAN GENUINE VISUAL GROUNDING.

Model	Normal	Visual Evidence Degradation					Visual Source Consistency	Task-Relevant Scene Variation				Camera $\uparrow$ Staleness	Overall Avg.	
		Target Obj. $\diamond$	Target Rec. $\diamond$	Interact. $\diamond$	Temp. Obs. $\uparrow$	View $\uparrow$	Cross-View $\diamond$	Task-Rel. $\uparrow$	Task Prec. $\uparrow$	Visual State $\diamond$	Target $\uparrow$			
		Mask.	Mask.	Cue Mask.	Corr.	Unavail.	Contrad.	Distractors	Var.	Ambig.	Geom. Var.			
VLA	OpenVLA-OFT	97.0	91.28 $\downarrow$ 5.2	54.34 $\downarrow$ 42.7	16.23 $\downarrow$ 80.8	76.39 $\downarrow$ 20.6	29.25 $\downarrow$ 67.8	54.47 $\downarrow$ 42.5	65.08 $\downarrow$ 31.9	39.38 $\downarrow$ 57.6	88.22 $\downarrow$ 8.8	81.74 $\downarrow$ 15.3	24.02 $\downarrow$ 73.0	56.40 $\downarrow$ 40.6
	π_0	92.5	81.00 $\downarrow$ 11.5	87.50 $\downarrow$ 5.0	71.59 $\downarrow$ 20.9	70.94 $\downarrow$ 21.6	28.19 $\downarrow$ 64.3	51.53 $\downarrow$ 41.0	61.25 $\downarrow$ 31.2	37.13 $\downarrow$ 55.4	70.67 $\downarrow$ 21.8	79.31 $\downarrow$ 13.2	27.00 $\downarrow$ 65.5	60.55 $\downarrow$ 31.9
	$\pi_{0.5}$	97.2	<u>93.41</u> $\downarrow$ 3.8	<u>95.66</u> $\downarrow$ 1.5	47.54 $\downarrow$ 49.7	<u>82.80</u> $\downarrow$ 14.4	22.72 $\downarrow$ 74.5	59.21 $\downarrow$ 38.0	70.92 $\downarrow$ 26.3	45.50 $\downarrow$ 51.7	85.78 $\downarrow$ 11.4	85.35 $\downarrow$ 11.9	20.52 $\downarrow$ 76.7	64.49 $\downarrow$ 33.7
WAM	FastWAM	98.8	93.38 $\downarrow$ 5.4	95.09 $\downarrow$ 3.7	69.42 $\downarrow$ 29.4	81.34 $\downarrow$ 17.5	10.12 $\downarrow$ 88.7	<u>63.63</u> $\downarrow$ 35.2	69.50 $\downarrow$ 29.3	<u>43.63</u> $\downarrow$ 55.2	94.67 $\downarrow$ 4.1	88.24 $\downarrow$ 10.6	17.00 $\downarrow$ 81.8	<u>66.00</u> $\downarrow$ 32.8
	LingBot-VA	97.0	84.69 $\downarrow$ 12.3	96.78 $\downarrow$ 0.2	<u>70.43</u> $\downarrow$ 26.6	67.89 $\downarrow$ 29.1	21.00 $\downarrow$ 76.0	70.11 $\downarrow$ 26.9	<u>69.75</u> $\downarrow$ 27.2	45.50 $\downarrow$ 51.5	<u>93.33</u> $\downarrow$ 3.7	86.39 $\downarrow$ 10.6	34.50 $\downarrow$ 62.5	67.31 $\downarrow$ 29.7
	LaWAM	98.5	93.44 $\downarrow$ 5.1	91.06 $\downarrow$ 7.4	69.13 $\downarrow$ 29.4	85.97 $\downarrow$ 12.5	17.97 $\downarrow$ 80.5	58.37 $\downarrow$ 40.1	69.50 $\downarrow$ 29.0	41.38 $\downarrow$ 57.1	91.33 $\downarrow$ 7.2	<u>87.75</u> $\downarrow$ 10.8	16.60 $\downarrow$ 81.9	65.68 $\downarrow$ 32.8